Genetically Engineering *Danio rerio* to Express PETase and MHETase Enzymes for Combating Microplastic Accumulation

BASIS Independent Silicon Valley A

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Problem and Proposed Solution

There are over 24.4 trillion pieces of microplastic (<5mm) littering our oceans [1] and the consumption of these microplastics is becoming a serious issue in marine organisms. Seventy-three percent of fish tested in Canada, 95% of fish tested off of the coast of Taiwan, and 49% of fish across 587 global species have been found with microplastic in their tissues. Polyethylene terephthalate (PET) is the fourth most common plastic pollutant in the world and regularly ends up being consumed by marine organisms. These PET microplastics are notorious for spreading through food chains and eventually making their way to our plates when we eat fish. In humans, microplastics leach phthalates and antimony, two potential carcinogens. Microplastics in humans have also been linked to oxidative stress, neurotoxicity, and inflammation. [2] Thus, developing a method to prevent microplastic accumulation in marine organisms is critical to protect humans and save lives. In this work, we propose the integration of the plastic degrading PETase and MHETase genes from the marine bacteria *Ideonella sakaiensis* into the fish model organism *Danio rerio.* These genes when integrated with a gut specific promoter into the fish will allow for the degradation of PET microplastics into non-toxic compounds in the fish gut, thus preventing microplastic spread into humans.

History and Alternatives

PET polymers were first synthesized in 1941 by John Rex Whinfield. In addition to packaging, PET plastics are regularly used in clothing textiles like polyester and to develop thermoplastic resins. In 2016, PET plastic production exceeded 56 million tons with around a $3.5 billion market size. In addition to PET, other common plastics include polyethylene, polypropylene and polyvinyl chloride plastics.

The PETase and MHETase enzymes were discovered in the bacterium *Idonella. sakaiensis* by Japanese researchers studying a plastic disposal site. Transgenic *E. coli* and micro-algae have been modified to express these enzymes, but a multicellular host is yet to be explored.

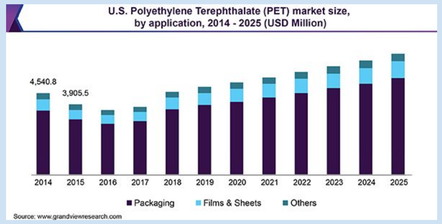

Figure 1: Market size of PET plastics

Currently, there are very limited options in preventing the accumulation of microplastics in fish other than employing greater conservation efforts to prevent plastic waste from entering the wild or raising farmed fish with more care in cleaner plastic-free environments.

Economic Feasibility and Cost Analysis

Treatment	*I. sakaiensis*	PETase expressing *S. abundans*	Fish (*D. rerio*)
Byproduct	None	Ethylene glycol, Terephthalic acid	None
Effect	None	Algal blooms leading to oxygen deprivation and death of other organisms.	N/A
Efficiency	1 PET wafer/60 weeks	59-87% efficiency, 1 day, 90,000 particles of PM, PMMA, or PLA per mL, 2 µm wide	Projected to be better than *S. abundans*
Cost	$300/NCTC ampule	*$125/culture	Projected to be $25/fish
Permanence	Permanent (landfill)	Permanent (ocean)	One time (ocean)

*Costs do not include the money spent to raise and care for the organisms

Detailed Solution and Sketches

The Tol-2 transposase system will be utilized to design to deliver the PETase and MHETase genes into the zebrafish genome. Both of these genes will be driven by the intestinal fatty acid binding protein (I-FABP) gut specific promoter. The Tol-2 transposase will be expressed directly from the same plasmid, and will integrate the genes found between the ITRs in the plasmid.

1. **Gene Isolation:** First the PETase and MHETase genes will be PCR amplified from *I. sakiensis* genomic DNA with the appropriate sticky ends.
2. **Promoter Isolation:** Similarly, the I-FABP gut specific promoter will be PCR amplified from zebrafish genomic DNA.
3. **Backbone Preparation:** The transposase plasmid backbone obtained from Addgene will be digested with the appropriate restriction enzymes to linearize it and allow for the insertion of the gene fragments.
4. **Cloning:** Gibson cloning will be performed with T4 DNA ligase to join the promoter sequence, PETase and MHETase into the transposase backbone with the overlapping sticky ends.
5. **Transformation:**The transposon plasmid will be electroporated of chemically transformed into competent DH5α *E. coli.*
6. **Microinjection:** The plasmid will be extracted through alkaline lysis and transposon DNA will be micro-injected into fertilized zebrafish embryos.

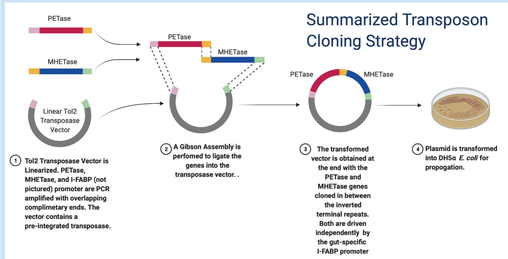

Figure 2: Transposon Cloning Strategy

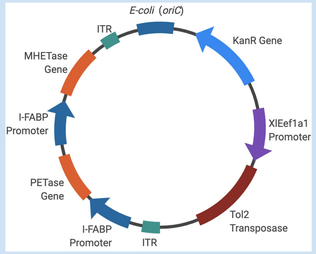

Figure 3: Theoretical Plasmid Construction

Testing and Benchmarking

The microinjection would first be tested in zebrafish, *D. rerio*, to confirm if the gene can be successfully integrated, expressed, and function in fish.

For the experimental group, 400 fish would be microinjected, and then 50 fish will be placed into a 500 L container. Each container will contain conditioned tap water with 0.0 µg/L, 0.0361 µg/L, 0.108 µg/L, and 0.181 µg/L PET. The PET will be 10 µm wide beads . To maintain the plastic concentration, the water would be replaced with new water with the same amount of plastic every 2 weeks. A sample of the cleaned out water would be tested for its microplastic concentration using Fourier-transform infrared (FTIR) spectroscopy, which was chosen over other imaging technology since it reduces overestimates of potential microplastic count.

At specific time intervals (10 days, 7 weeks, 14 weeks)[6], 33 fish will be euthanized by freezing on ice and parts of the fish gut, gill, and muscle would be removed to test the success of gene integration by qPCR. If that is successful, the expression of MHETase and PETase will be tested through SDS-PAGE and Western Blot. Parts of the fish mentioned above will also be used with FTIR to determine the presence of microplastics in the fish.

For the control group, 400 fish without the transposon would be raised and then put through the same conditions as the experimental group. The concentrations of plastic, MHETase, and PETase in the fish would be tested and compared against the experimental group. If we are successful in expressing a functional MHETase and PETase gene in the fish, then we will test the stability of the gene by raising the micro-injected fish for 3 generations in the same conditions stated above.

Limitations and Technical Challenges

A few limitations we came across included using store-bought fishmeal for the zebrafish diet. Fishmeal can be affected by PET buildup in the oceans, which may skew the results of testing, provided that our fish are not supposed to be exposed PET in the fishmeal [3]. Also, our target organism is the Atlantic salmon (*Salmo salar*), which are saltwater fish, while our model organism, the zebrafish, are freshwater fish. Thus due to this disparity, more work will be needed to translate our proposed laboratory model to the real world.

The PETase enzyme itself also has an optimal temperature ranging from 25-45°C, though it may still remains slightly less active at other temperatures near the range [4]. For *I. sakaiensis*, the PETase enzyme can effectively function at temperatures as low as 15°C, but this is due to the presence of AFP antifreeze proteins present in bacteria and archaea [5] that help maintain protein stability. These antifreeze proteins are not present in Atlantic salmon, which may slightly reduce the PETase's efficiency to degrade PET. As for technical challenges, fish survival (until adulthood) rates before exposing them to PET may be negatively affected by the microinjection or the integration of the PETase and MHETase genes, though it that is another variable to investigate.

Ethical and Societal Concerns

The principal ethical concerns that arises from this work is whether the integration of the PETase and MHETase genes has any adverse environmental effects, toxicity to humans or pose any dangers transgenic fish itself.

PETase and MHETase are not known to have short-term detrimental environmental effects. However, the fact that these enzymes are naturally found in the wild further contributed to the argument that they are safe for use in the long-term.

PETase and MHETase expression in fish may be toxic however. Transposon integrations are random and it is possible the PETase and MHETase genes land in genomic regions that when disrupted are detrimental to fish health. It is also possible that the genes are overexpressed or interfere with other proteins in the fish, leading to downstream toxicity. However, this is agan unlikely and toxicity will be screened for in preliminary experimentation.

Some people are skeptical about GMO foods in general, as these foods are rather new, and there is no research about long term effects. There is also the concern of having the transposase gene be integrated in their own human genome, but without a Tol2 transposase being expressed in a human, it is highly unlikely that such an event will occur.

Overall, this work has the potential to protect the environment and save countless lives. The countless benefits these genetically modified fish provide clearly outweigh potential risks of this work.

References and Acknowledgements

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